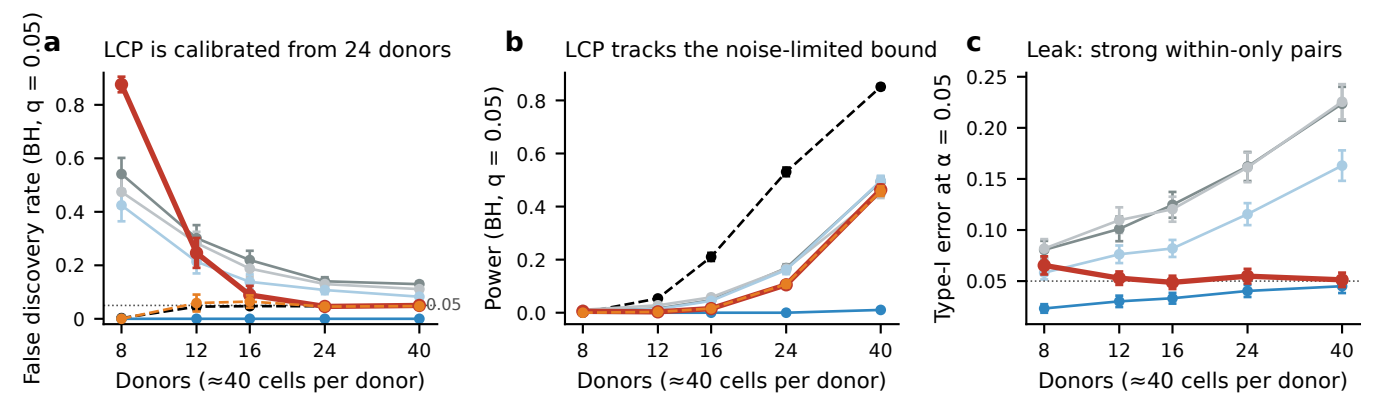
a
LCP is calibrated from 24 donors
False discovery rate (BH, q = 0.05)
0.8
0.6
0.4
0.2
0
0.05
8
12
16
24
40
Donors (≈40 cells per donor)
b
LCP tracks the noise-limited bound
Power (BH, q = 0.05)
0.8
0.6
0.4
0.2
0.0
8
12
16
24
40
Donors (≈40 cells per donor)
c
Leak: strong within-only pairs
Type-I error at α = 0.05
0.25
0.20
0.15
0.10
0.05
8
12
16
24
40
Donors (≈40 cells per donor)
Oracle: true donor means
Noise-limited oracle (bound)
LCP (this work)
Pseudobulk Pearson
Count-scale means, uncorrected
Standardized log means
Moment-corrected + jackknife